Probing the Deep: Exoplanet Potential Fields

Investigating Gravity Moments and Magnetic Dynamos

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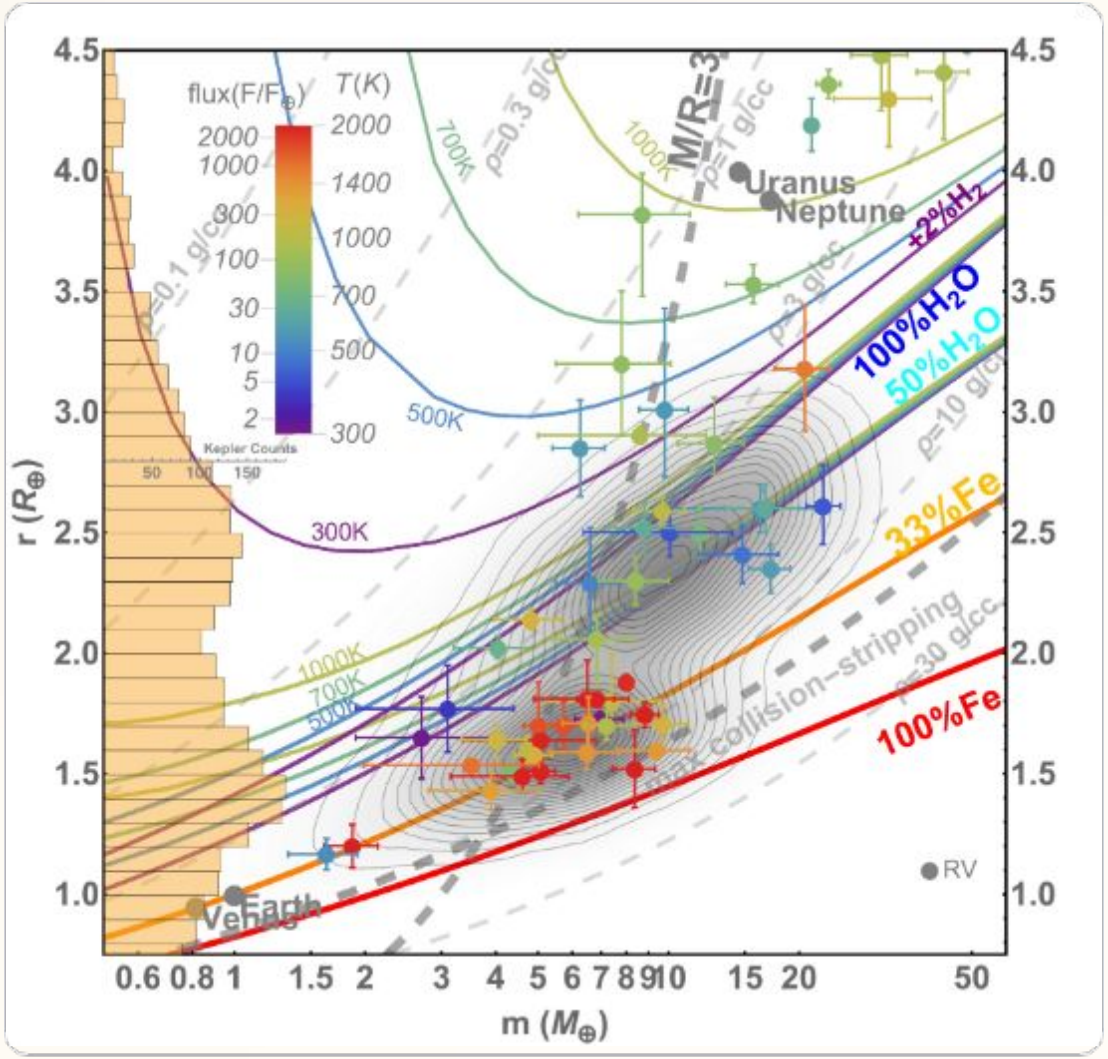
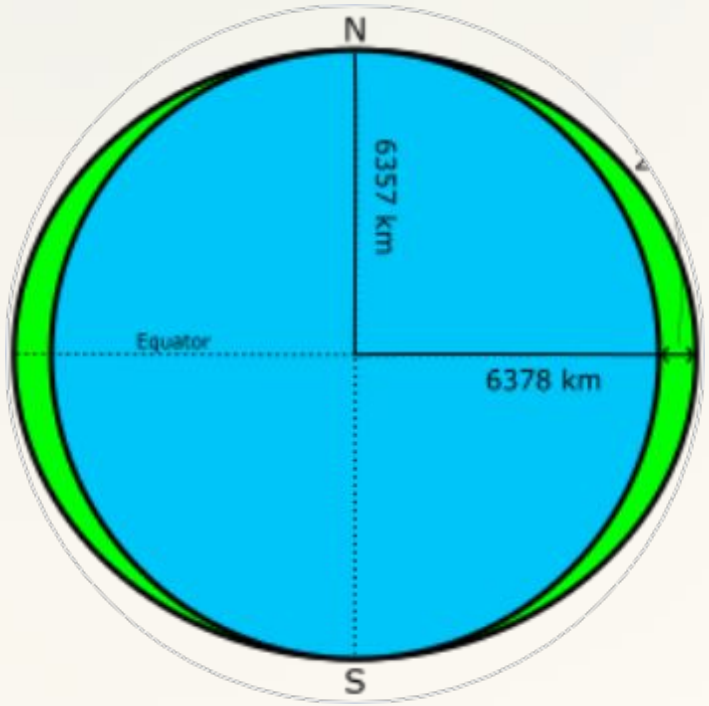
The Interior Characterization Challenge

Characterizing exoplanets currently relies on **Mass** and **Radius**, which creates a fundamental "degeneracy."

- Rocky planets with iron cores can mimic water-worlds.
- Bulk density alone cannot distinguish composition.
- Potential fields offer a direct probe into internal mass distribution.



Rotational Bulge and J2 Moments



Fast-rotating exoplanets exhibit **Rotational Oblateness**. This "bulge" at the equator perturbs the gravitational potential field.

$$V_g = -\frac{GM}{r}\left(1 - J_2 \frac{R_\oplus^{\,2}}{r} P_2(\cos\theta)\right)$$

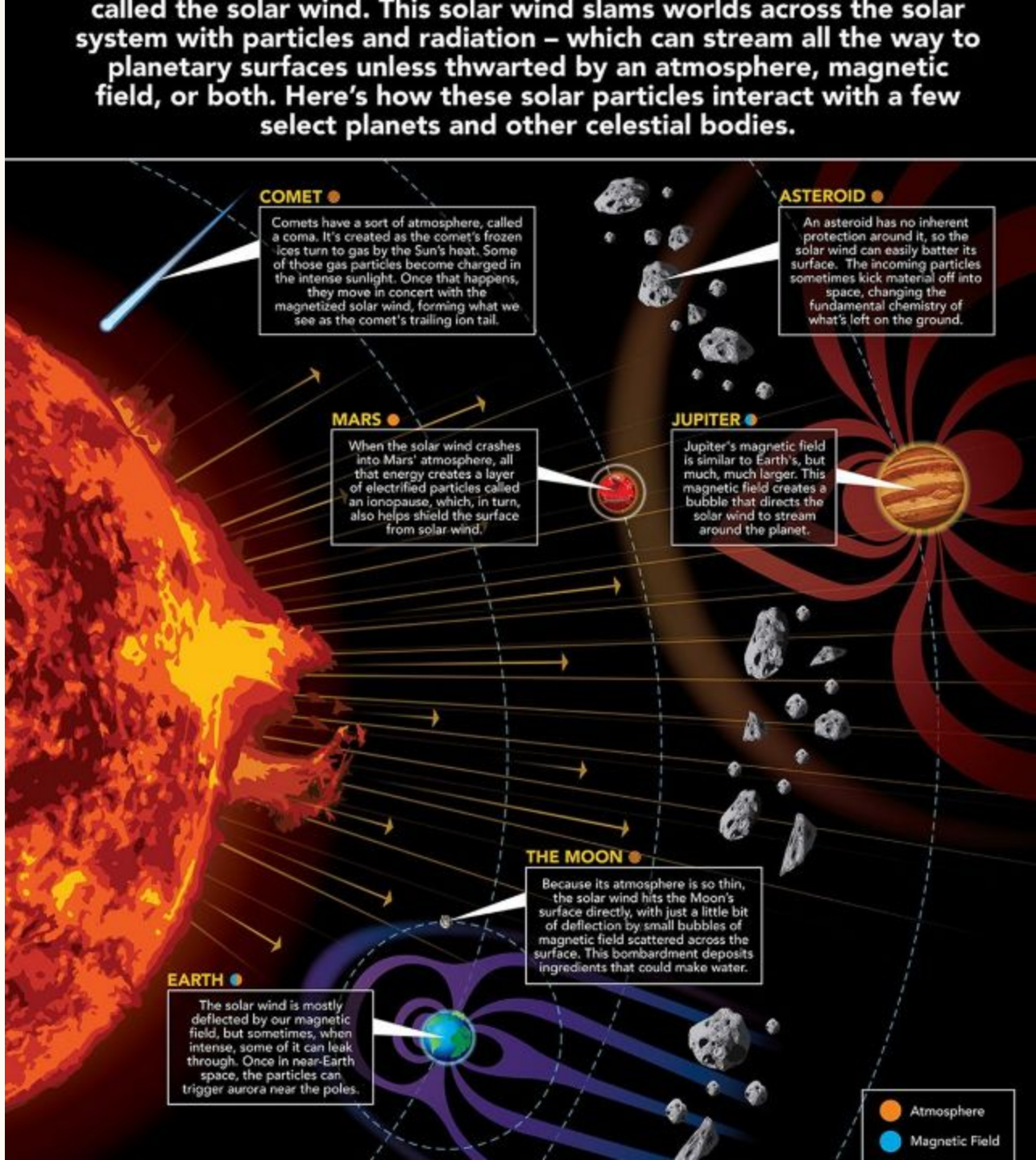
Measuring J_2 allows us to constrain the core mass fraction and density gradients.

Tidal Potential Probes

Magnetism and Habitability

A global magnetic field indicates a **liquid, convecting core** (Geodynamo). It is a vital shield against stellar winds.

- Many "habitable" planets are tidally locked.
- Slow rotation prevents strong Coriolis force.
- Research shows 50% of CHZ exoplanets lack this shield.



Observational Precision Requirements

Facility	White Noise (ppm)	Transits Required (J2 Detection)
JWST (NIRSpec)	23	~2
PLATO	63	~10
CHEOPS	71	~13
TESS	360	>300

Signals from shape deformation are subtle (~20-40 ppm), requiring ultra-high precision missions.

Current Research Methods

Forward Modeling

Constructing analytical 3D models of deformed ellipsoids and projecting them onto stellar disks to simulate expected flux drops during transits.

Bayesian Inference

Using MCMC (Markov Chain Monte Carlo) algorithms to fit parameters like the Love number k_2 against actual light curves.

Complexity and Confusion

"Disentangling the parts-per-million signal of planetary shape from stellar limb-darkening noise remains the primary computational bottleneck."

— Synthesis of Current Literature

Current models are slow, requiring weeks of computation for a single system, and often fail to distinguish between different noise sources effectively.

The Target Frontier

20

Parts-Per-Million

This is the required sensitivity threshold. Our goal is to extract these minuscule "wiggles" in light curves that reveal the presence of iron cores and liquid interiors.

Solution: Dual-Input CNN

Synthesis

Generating millions of synthetic light curves with randomized physical perturbations to provide ground-truth labels for deep learning.

Architecture

Developing a 1D-CNN branch for raw time-series data and a Dense branch for scalar system parameters (mass, star type).

Validation

Predicting potential field coefficients in seconds, bypassing the statistical bottlenecks of traditional MCMC methods.

Research Roadmap (11 Months)

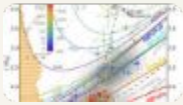
Phase 1: Synthesis	Phase 2: Dev	Phase 3: Training	Phase 4: Launch
Months 1-3: Forward modeling and pipeline engineering for synthetic data.	Months 4-6: Architecture design in PyTorch and model integration.	Months 7-9: Initial training and hyperparameter optimization.	Months 10-11: Validation against archival data and final reporting.

Questions?

Thank you for your attention.

Investigating Exoplanet Potential Fields via Deep Learning

Image Sources



https://lweb.cfa.harvard.edu/~lzeng/plots/mass_radius_20190518_HarpsN_only.jpg

Source: lweb.cfa.harvard.edu

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Source: ai-solutions.com

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Source: science.nasa.gov